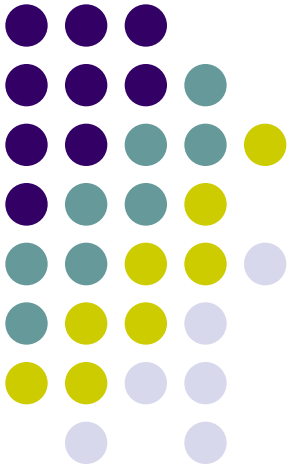
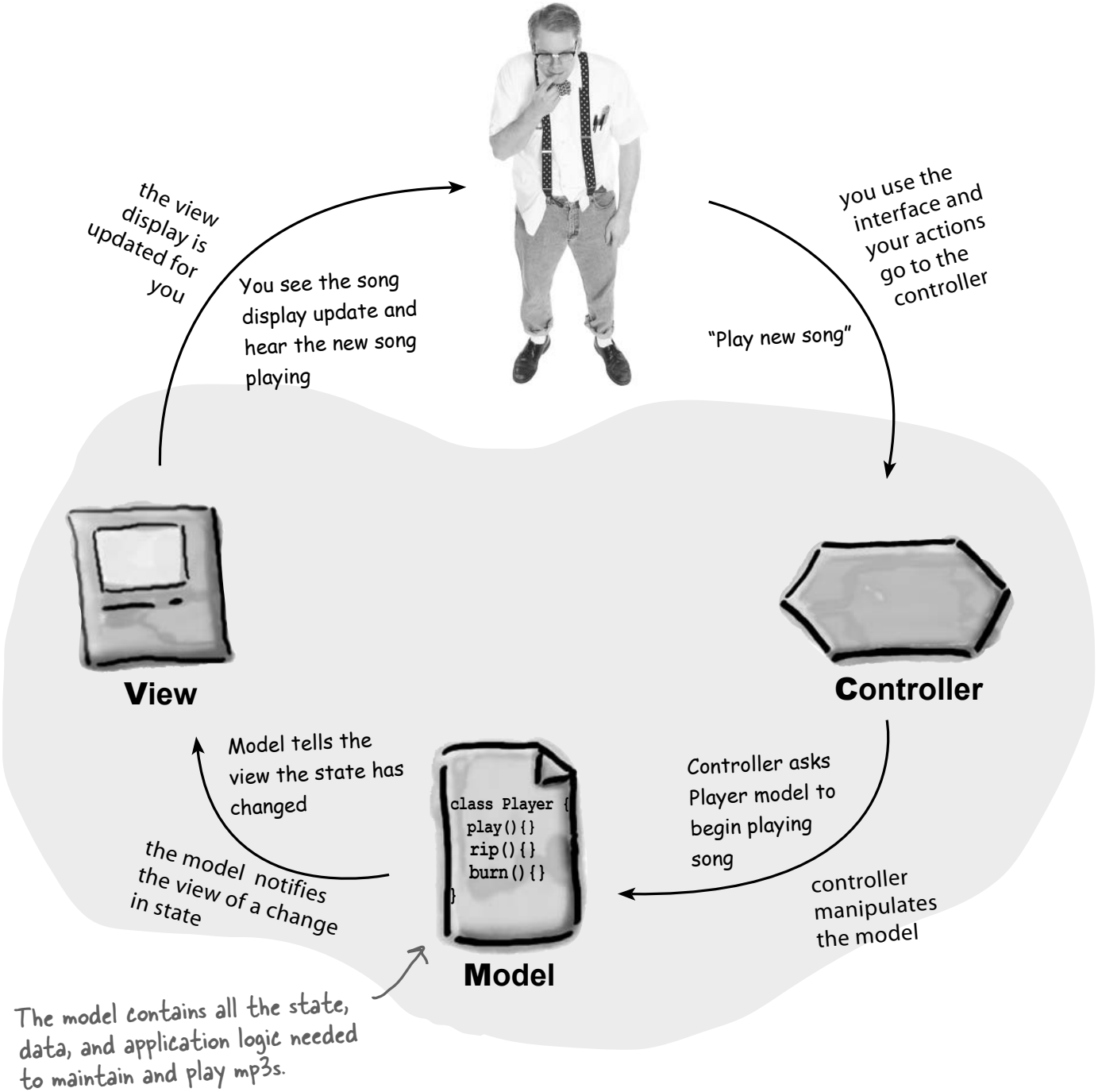
MVC

Model-View-Controller




the view display is updated for you
You see the song display update and hear the new song playing
you use the interface and your actions go to the controller
"Play new song"
View
Controller
Model tells the view the state has changed
class Player {
play(){}
rip(){}
burn(){}
}
Controller asks Player model to begin playing song
the model notifies the view of a change in state
controller manipulates the model
Model
The model contains all the state, data, and application logic needed to maintain and play mp3s.

CONTROLLER

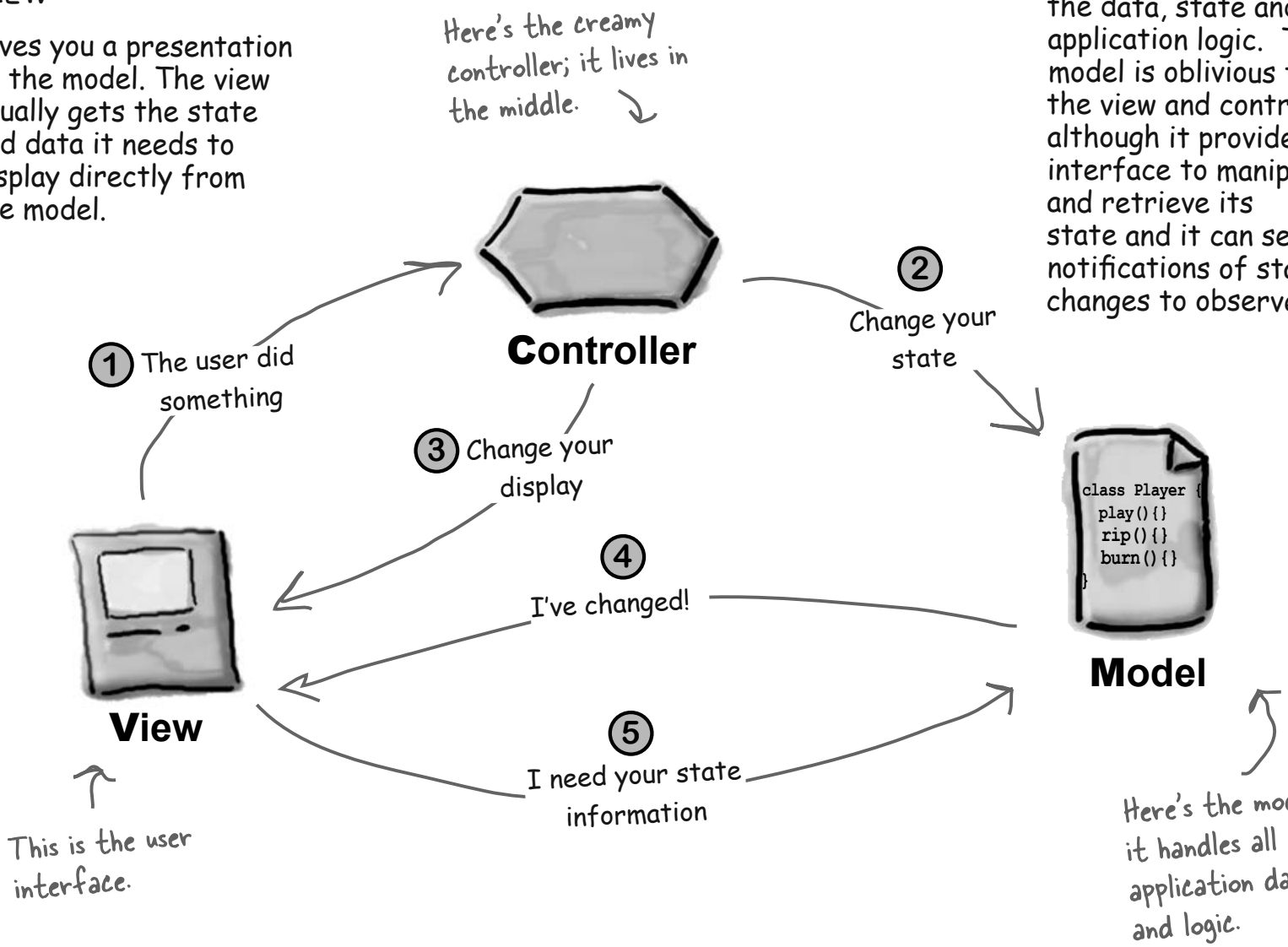
Takes user input and figures out what it means to the model.

MODEL

The model holds all the data, state and application logic. The model is oblivious to the view and controller, although it provides an interface to manipulate and retrieve its state and it can send notifications of state changes to observers.

VIEW

Gives you a presentation of the model. The view usually gets the state and data it needs to display directly from the model.



1. **You're the user — you interact with the view.**
The view is your window to the model. When you do something to the view (like click the Play button) then the view tells the controller what you did. It's the controller's job to handle that.

2. **The controller asks the model to change its state.**
The controller takes your actions and interprets them. If you click on a button, it's the controller's job to figure out what that means and how the model should be manipulated based on that action.

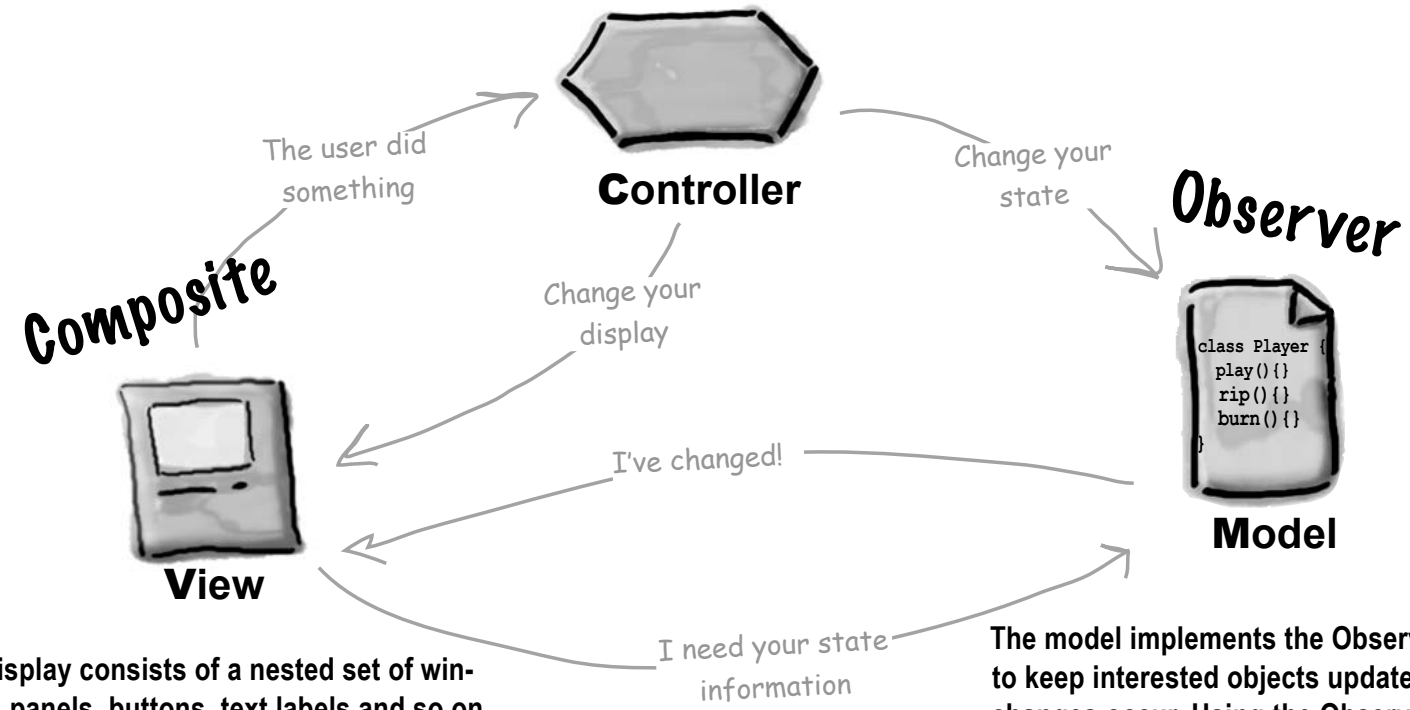
3. **The controller may also ask the view to change.**
When the controller receives an action from the view, it may need to tell the view to change as a result. For example, the controller could enable or disable certain buttons or menu items in the interface.

4. **The model notifies the view when its state has changed.**
When something changes in the model, based either on some action you took (like clicking a button) or some other internal change (like the next song in the playlist has started), the model notifies the view that its state has changed.

5. **The view asks the model for state.**
The view gets the state it displays directly from the model. For instance, when the model notifies the view that a new song has started playing, the view requests the song name from the model and displays it. The view might also ask the model for state as the result of the controller requesting some change in the view.

Strategy

The view and controller implement the classic Strategy Pattern: the view is an object that is configured with a strategy. The controller provides the strategy. The view is concerned only with the visual aspects of the application, and delegates to the controller for any decisions about the interface behavior. Using the Strategy Pattern also keeps the view decoupled from the model because it is the controller that is responsible for interacting with the model to carry out user requests. The view knows nothing about how this gets done.



The display consists of a nested set of windows, panels, buttons, text labels and so on. Each display component is a composite (like a window) or a leaf (like a button). When the controller tells the view to update, it only has to tell the top view component, and Composite takes care of the rest.

The model implements the Observer Pattern to keep interested objects updated when state changes occur. Using the Observer Pattern keeps the model completely independent of the views and controllers. It allows us to use different views with the same model, or even use multiple views at once.

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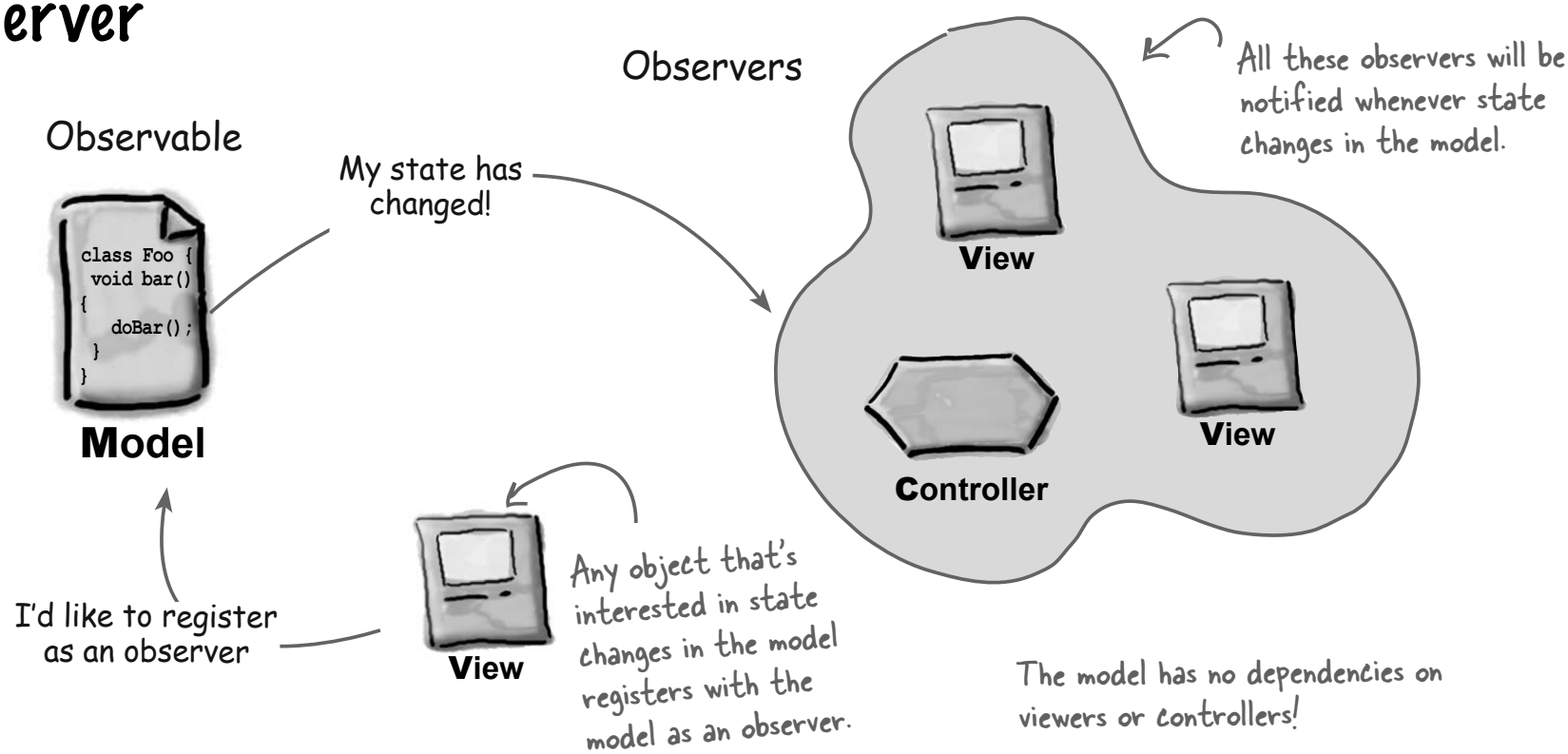
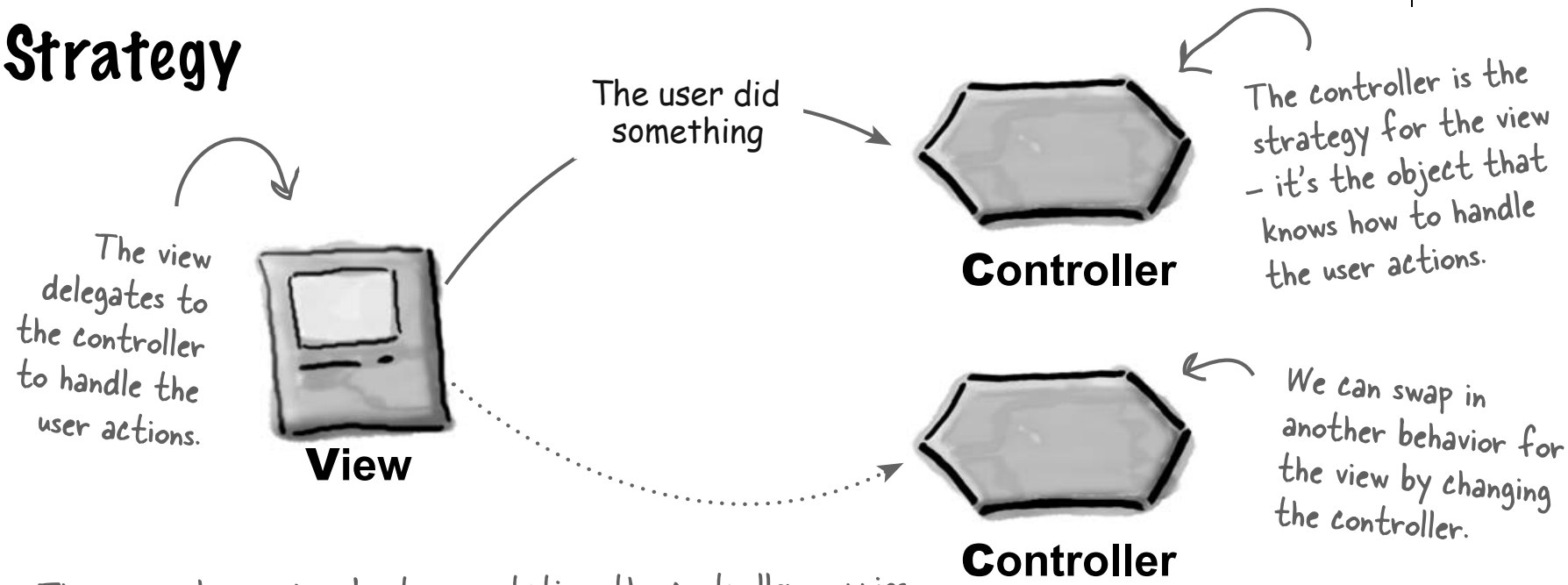
Observer

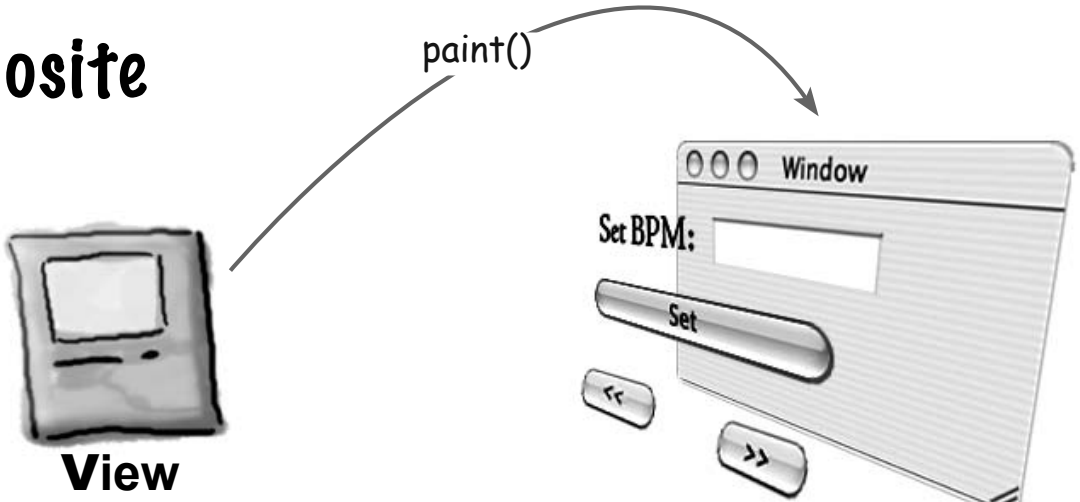

Observers
Observable
My state has changed!
class Foo {
void bar()
{
doBar();
}
}
Model
View
Controller
View
All these observers will be notified whenever state changes in the model.
I'd like to register as an observer
View
Any object that's interested in state changes in the model registers with the model as an observer.
The model has no dependencies on viewers or controllers!

Strategy



The view only worries about presentation, the controller worries about translating user input to actions on the model.

Composite



The view is a composite of GUI components (labels, buttons, text entry, etc.). The top level component contains other components, which contain other components and so on until you get to the leaf nodes.

Reviews

- The Model View Controller Pattern (MVC) is a compound pattern consisting of the Observer, Strategy and Composite patterns.
- The model makes use of the Observer Pattern so that it can keep observers updated yet stay decoupled from them.
- The controller is the strategy for the view. The view can use different implementations of the controller to get different behavior.

- The view uses the Composite Pattern to implement the user interface, which usually consists of nested components like panels, frames and buttons.
- The patterns work together to decouple the three players in the MVC model, which keeps designs clear and flexible.